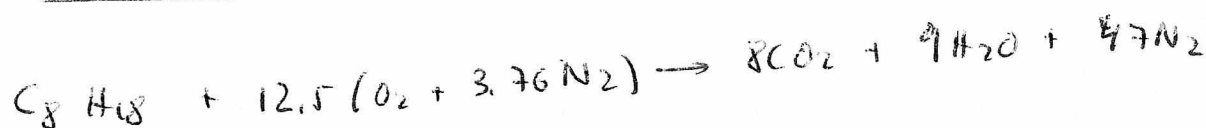


Problema - 1

Hallar la temperatura de combustión adiabática para la reacción estequiométrica de la gasolina líquida (C_8H_{18}) cuando se usa como comburente aire y oxígeno puro. En ambos casos el comburente se suministra a $25^\circ C$.

	C_p [kJ/kmol-K]		$\bar{h}_f^0$ [kJ/kmol]
CO_2	58.4	CO_2	-393520
H_2O	47.15	H_2O (v)	-241820
N_2	34.9	C_8H_{18} (l)	-249910

a) comburente aire



$$\sum n_p \bar{h}_p = \sum n_r \bar{h}_r$$

$$8[\bar{h}_{fCO_2}] + 9[\bar{h}_{fH_2O}] - \bar{h}_{fC_8H_{18}} + [8\bar{C}_{pCO_2} + 9\bar{C}_{pH_2O} + 47\bar{C}_{pN_2}] \times [T_{ca} - T_0] = 0$$

$$\bar{h}_{comb}^0 + [8\bar{C}_{pCO_2} + 9\bar{C}_{pH_2O} + 47\bar{C}_{pN_2}] (T_{ca} - T_0) = 0$$

$$T_{ca} = T_0 + \frac{\bar{h}_{comb}^0}{8\bar{C}_{pCO_2} + 9\bar{C}_{pH_2O} + 47\bar{C}_{pN_2}}$$

$$\begin{aligned} \bar{h}_{comb}^0 &= -8 \times \frac{393520}{249910} - 9 \times 241820 + 249910 = \\ &= -5074630 \text{ kJ/kmol} \end{aligned}$$

$$T_{ca} = 298 + \frac{5074630}{8 \times 58.4 + 9 \times 47.15 + 47 \times 34.9} =$$

$$= 2302.32 \text{ K} = \underline{\underline{2029.32^\circ C}}$$

2.

b) comburente O₂



$$\underline{\underline{T_{ca}}} = 298 + \frac{5074630}{8 \times 58,4 + 9 \times 47,15 + 0} =$$

$$= 5989,92 \text{ K} = \underline{\underline{5716,92^\circ\text{C}}}$$